

■ EXAMPLE 2.14

Suppose that a zero-mean stationary stochastic process has the autocorrelation function

$$R_X(\tau) = \sigma^2 e^{-\beta|\tau|} \quad (2.101)$$

where β is a positive real number. The power spectrum is computed from Equation (2.92) as

$$\begin{aligned} S_X(\omega) &= \int_{-\infty}^{\infty} \sigma^2 e^{-\beta|\tau|} e^{-j\omega\tau} d\tau \\ &= \int_{-\infty}^0 \sigma^2 e^{(\beta-j\omega)\tau} d\tau + \int_0^{\infty} \sigma^2 e^{-(\beta+j\omega)\tau} d\tau \\ &= \frac{\sigma^2}{\beta-j\omega} + \frac{\sigma^2}{\beta+j\omega} \\ &= \frac{2\sigma^2\beta}{\omega^2 + \beta^2} \end{aligned} \quad (2.102)$$

The variance of the stochastic process is computed as

$$\begin{aligned} E[X^2(t)] &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2\sigma^2\beta}{\omega^2 + \beta^2} d\omega \\ &= \frac{\sigma^2\beta}{\pi} \left[\frac{1}{\beta} \tan^{-1} \frac{\omega}{\beta} \right]_{-\infty}^{\infty} \\ &= \sigma^2 \\ &= R_X(0) \end{aligned} \quad (2.103)$$

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2.7 SIMULATING CORRELATED NOISE

In optimal filtering research and experiments, we often have to simulate correlated white noise. That is, we need to create random vectors whose elements are correlated with each other according to some predefined covariance matrix. In this section, we will present one way of accomplishing this.

Suppose we want to generate an n -element random vector w that has zero mean and covariance Q :

$$Q = \begin{bmatrix} \sigma_1^2 & \cdots & \sigma_{1n} \\ \vdots & & \vdots \\ \sigma_{1n} & \cdots & \sigma_n^2 \end{bmatrix} \quad (2.104)$$

Since Q is a covariance matrix, we know that all of its eigenvalues are real and nonnegative. We can therefore denote its eigenvalues as μ_k^2 :

$$\lambda(Q) = \mu_k^2 \quad (k = 1, \dots, n) \quad (2.105)$$

Suppose the eigenvectors of Q are found to be $d_1, \dots, d_n$. Augment the d_i vectors together to obtain an $n \times n$ matrix D . Since Q is symmetric, we can always choose